

Calculus III
Fall, 2024

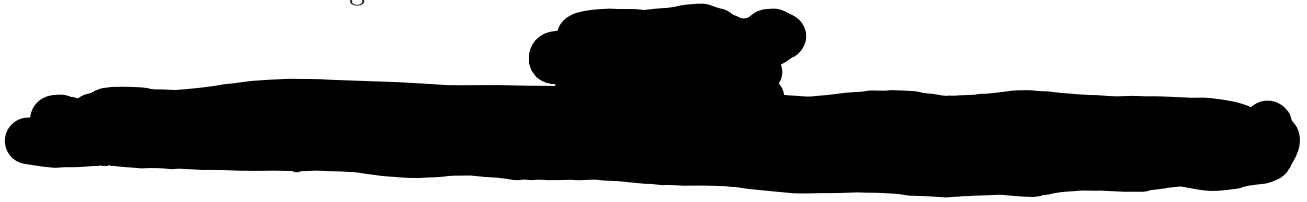
Final Exam

Name: _____
Dr. B.

Part I (200 points) Show work for credit. Calculator OK (but not useful).

(18pts) 1. Consider the line  plane

(18^{pts}) 2. Show that the following limit



- (18^{pts}) 3. Suppose a mountain can be represented by the function [REDACTED]
- (a) (9 pts) If you're on the mountain at the point $(0, 1, 0)$, in what direction should you [REDACTED]

- (b) (9 pts) If you're at the point [REDACTED]
what is the rate of change of elevation as you start your walk?

(18^{pts}) 4. Give an equation for the tangent plane to the surface



(20^{pts}) 5. Find all absolute maxima and minima of the function

[REDACTED]

(18pts)

6. Evaluate $\int \int \int_E$

(18pts)

7. Evaluate $\int_C$ 

(18pts)

8. Compute $\int_C \text{[redacted]} dx + \text{[redacted]} dy$, where C is the curve [redacted] .

- (18^{pts}) 9. Let C be the radius 5 circle centered at the origin traversed counter-clockwise. Compute $\int_C \text{[redacted]} dx + \text{[redacted]} dy$. [redacted]

(18^{pts}) **10.** Let S be the surface parameterized by

Compute $\iint_S$ dS .

(18^{pts}) 11. Compute

$$\int \int_S \mathbf{F} \cdot d\mathbf{S},$$
